

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	24-08-26

Code of Conduct

I, T Bhanu Teja/192472259 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bhanu Teja

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement

Inefficient parsing for long queries

Example user query:

“Show me the transactions with the card from last month”

a) Analyze the ambiguity present in the given sentence using CFG

b) Evaluate the limitations of CFG in handling real-world conversational queries

c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing

d) Justify how your proposed method improves accuracy and efficiency in an industry setting

ANSWER:

Given sentence:

“Show me the transactions with the card from last month.”

a) Analyze the ambiguity using CFG

A Context-Free Grammar (CFG) represents a sentence using grammatical rules such as:

The given sentence contains a **prepositional phrase (PP) attachment ambiguity**.

The phrase:

“with the card”

can be interpreted in two ways.

Interpretation 1: Card is associated with the transactions

Show

```
└── transactions
    └── with the card
        └── from last month
```

Meaning:

Show transactions that were made **using the card** during last month.

Interpretation 2: Card is associated differently with the action/query

The PP can potentially attach to a higher phrase in the parse structure, producing a different interpretation.

A simplified ambiguity is:

```
      VP
     /  \
    V    NP
    |
transactions
  /    \
PP with PP from
the card last month
```

versus another possible attachment:

```
      VP
     /  \
    V    NP
    |
transactions
  \
   PP
with the card
  \
   PP from
```

last month

The ambiguity arises because ordinary CFG rules describe **syntactic structure**, but do not inherently provide enough information to determine the intended meaning.

b) Evaluate the limitations of CFG in real-world conversational queries

CFG is useful for representing basic grammatical structures, but it has several limitations in banking chatbots.

1. Difficulty handling ambiguity

CFG may generate multiple valid parse trees without knowing which interpretation is intended.

2. No probability

A basic CFG does not assign probabilities to competing parse trees. Therefore, it cannot easily select the most likely interpretation.

3. Weak agreement handling

Rules such as:

do not inherently enforce subject-verb agreement.

4. Poor handling of conversational language

Real users may provide incomplete queries such as:

“Transactions last month?”

or:

“Show card transactions.”

A strict CFG may fail to parse such inputs.

5. Inefficient for long sentences

Large grammars can produce many possible derivations, increasing computational cost.

6. Limited semantic information

CFG primarily represents **syntax**, while a banking chatbot needs to understand entities such as:

- transaction
- card
- date
- amount
- account

Therefore, CFG alone is insufficient for a production-level conversational banking system.

c) Propose an improved solution using PCFG, feature structures, or Earley parsing

A **hybrid approach** combining PCFG, feature structures, and Earley parsing is more suitable.

Proposed architecture

User Query



Tokenization



CFG / Earley Parser



Feature Structure Analysis



Multiple Parse Trees



PCFG Probability Ranking



Semantic Interpretation



Banking Intent + Entities



Database Query



Response

PCFG

A Probabilistic CFG assigns probabilities to grammar rules.

For example:

When multiple parse trees are possible, the system can select the **highest-probability interpretation**.

Feature structures

Features can represent information such as:

NP:

NUMBER = plural

TYPE = transaction

For example:

Card:

TYPE = payment_card

Last month:

TIME = previous_month

Earley parsing

Earley parsing is useful because it can efficiently handle:

- ambiguous grammars;
- recursive structures;
- incomplete inputs;
- long sentences.

Thus, combining these techniques provides both **syntactic flexibility and probabilistic ambiguity resolution**.

d) Justify how the proposed method improves accuracy and efficiency

The hybrid approach improves the banking chatbot in several ways.

Problem	Proposed Solution
Ambiguous interpretation	PCFG ranking
Subject-verb agreement	Feature structures
Long/complex sentences	Earley parsing
Incomplete queries	Earley parsing
Semantic interpretation	Semantic analysis
Multiple possible parses	Probability-based selection

Result

CFG

↓

Structural analysis

Feature Structures

↓

Agreement + constraints

PCFG

↓

Most probable interpretation

Earley Parser

↓

Efficient parsing

Semantic Layer

↓

Banking Intent + Entities

Therefore, the proposed system provides **better accuracy, ambiguity resolution, scalability, and efficient processing** compared with using a basic CFG alone.

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response

Example command:

“Book a flight to Delhi with a window seat”

a) Analyze how the given sentence can lead to multiple parse structures

b) Analyze the limitations of top-down parsing in this real-time application

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

ANSWER:

Given command:

“Book a flight to Delhi with a window seat.”

a) Analyze how the sentence can lead to multiple parse structures

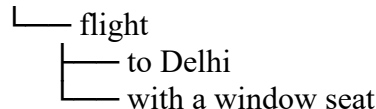
The command contains the phrase:

“with a window seat”

This phrase can have different syntactic attachments.

Interpretation 1 — Window seat is a property of the flight

Book



Meaning:

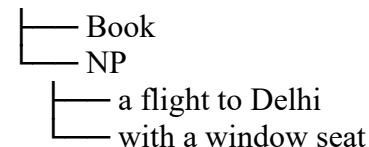
Book a flight to Delhi and request a window seat.

This is the most natural interpretation.

Interpretation 2 — PP attachment to another constituent

The phrase can syntactically attach at a higher level:

VP



The ambiguity comes from the fact that “with a window seat” is a prepositional phrase and can have more than one possible attachment.

A simple CFG can generate both structures without necessarily knowing which one is intended.

b) Analyze limitations of top-down parsing in real-time applications

Top-down parsing starts from the start symbol and predicts possible productions until it matches the input.

Problems

1. Backtracking

When a predicted rule does not match the input, the parser may need to backtrack.

Prediction

↓

Wrong choice

↓

Backtrack

↓

Try another rule

This increases processing time.

2. Repeated work

The parser may explore the same alternatives multiple times.

3. Poor ambiguity handling

Multiple possible interpretations may result in a large search space.

4. Incomplete speech input

Voice assistants often receive partial input:

“Book a flight to...”

A strict top-down parser may have difficulty making useful decisions before the entire sentence is available.

5. Real-time constraint

Voice assistants need fast responses, so excessive backtracking can reduce responsiveness.

c) Explain how Earley parsing improves ambiguity and partial-input handling

Earley parsing uses three main operations:

1. Prediction
2. Scanning
3. Completion

Earley processing

Input

↓

Prediction

↓

Scanning

↓

Completion

↓

Chart of partial states

↓

Possible parse structures

For example, when processing:

“Book a flight to Delhi...”

the parser can maintain partial states while waiting for the remaining input.

Earley parsing is useful because it:

- handles ambiguous CFGs;
- supports recursive grammar structures;
- stores partial parsing states;
- avoids unnecessary repeated work;
- can process incomplete input;
- is suitable for complex natural-language structures.

Therefore, Earley parsing is more flexible for voice-based applications.

d) Compare top-down and Earley parsing

Feature	Top-Down Parsing	Earley Parsing
Parsing direction	Start symbol → input	Chart-based
Ambiguity	Difficult	Handles ambiguity
Backtracking	Often required	Reduced through chart states
Partial input	Limited	Better support
Left recursion	Problematic for naïve versions	Can handle it
Long sentences	Can become inefficient	More robust

Feature	Top-Down Parsing	Earley Parsing
Real-time systems	Suitable for simple grammars	Better for complex input

Recommendation

For a large-scale voice assistant, Earley parsing is preferable when the grammar is complex and ambiguous. It can maintain partial interpretations and reduce unnecessary recomputation.

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization)

Real-time processing for hospital use

Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).

c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

ANSWER:

Given sentence:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

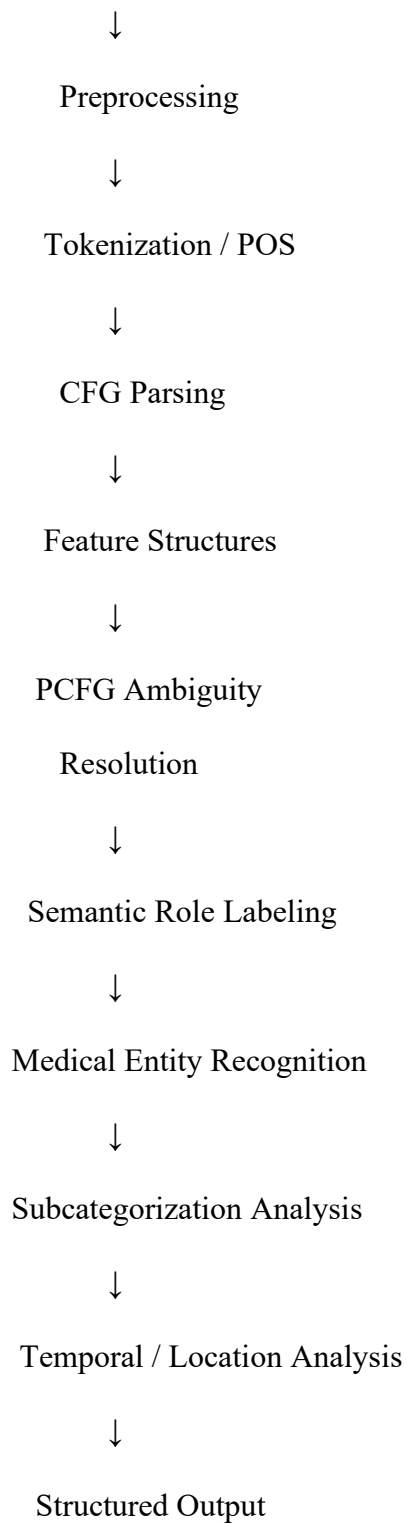
This is a complex sentence containing a **relative clause**, multiple actions, and a location.

a) Design a complete NLP architecture

The system should combine **CFG, PCFG, feature structures, semantic analysis, subcategorization, and efficient parsing.**

Proposed architecture

Medical Report



Role of each component

1. CFG

Provides the basic syntactic structure.

For example:

and the subject NP can contain a relative clause:

where RC represents the relative clause.

2. PCFG

If multiple syntactic interpretations exist, PCFG assigns probabilities to alternative parses and selects the most probable structure.

3. Feature structures

Feature structures can represent:

Doctor:

NUMBER = singular

PERSON = 3

ROLE = Agent

This helps enforce grammatical agreement.

4. Efficient parsing

Earley parsing or another chart-based parser can efficiently process the long and recursive sentence.

5. Semantic analysis

The system extracts relationships such as:

Doctor → recommends → medication

Doctor → schedules → follow-up visit

Follow-up visit → Location → Chennai

Doctor → reviewed → Patient

b) Step-by-step workflow from input to structured output

Step 1: Input

The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.

Step 2: Tokenization and POS tagging

Important words can be identified as:

doctor/NN

reviewed/VBD

patient/NN

week/NN

recommends/VBZ

starting/VBG

medication/NN

scheduling/VBG

visit/NN

Chennai/NNP

Step 3: Syntactic parsing

The sentence contains:

Main Clause:

The doctor ... recommends ...

Relative Clause:

who reviewed the patient last week

Coordinated Actions:

starting medication

+

scheduling a follow-up visit in Chennai

Step 4: Semantic role assignment

Doctor

↓

Agent

Patient



Patient/Theme

Medication



Treatment

Follow-up visit



Action

Chennai



Location

Step 5: Structured output

Diagnosis/Condition: Not explicitly stated

Agent: Doctor

Reviewed: Patient

Time: Last week

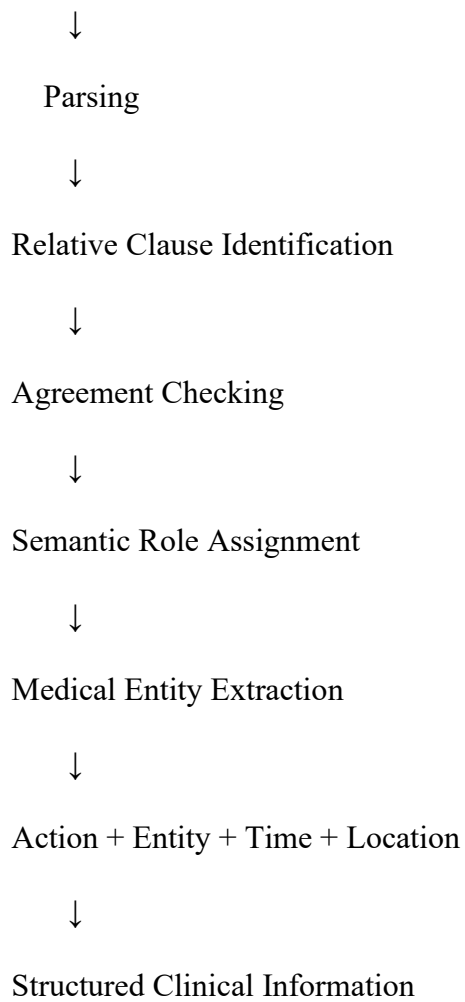
Action 1: Start medication

Action 2: Schedule follow-up visit

Location: Chennai

Overall workflow

Medical Sentence



c) Method to incorporate subcategorization, real-time processing and scalability

1. Incorporate subcategorization frames

Medical verbs impose restrictions on their complements.

For example:

recommend → NP / Gerund Clause

review → NP

schedule → NP

For the given sentence:

recommend

↓

starting medication

+

scheduling a follow-up visit

The system can validate whether the complements are grammatically compatible with the verb.

Similarly:

reviewed → NP

is satisfied by:

reviewed **the patient**

2. Handle real-time processing

A hospital system needs fast processing. The system can use:

- chart-based parsing;
- incremental parsing;
- caching of frequently used structures;
- efficient medical dictionaries;
- parallel processing;
- pre-trained medical language models.

Real-time architecture

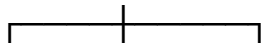
Incoming Report



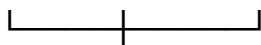
Incremental Parser



Parallel NLP Modules



POS NER Parsing



Semantic Analysis



Structured Result

3. Improve accuracy

Accuracy can be improved using:

- medical-domain training data;
- PCFG-based ambiguity resolution;
- feature agreement;
- medical Named Entity Recognition;
- semantic-role labeling;
- subcategorization frames;
- negation detection;
- temporal expression extraction.

For example, the system must distinguish:

“The patient is taking medication.”

from:

“The patient is not taking medication.”

Negation detection is therefore important in clinical NLP.

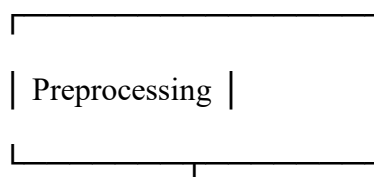
4. Improve scalability

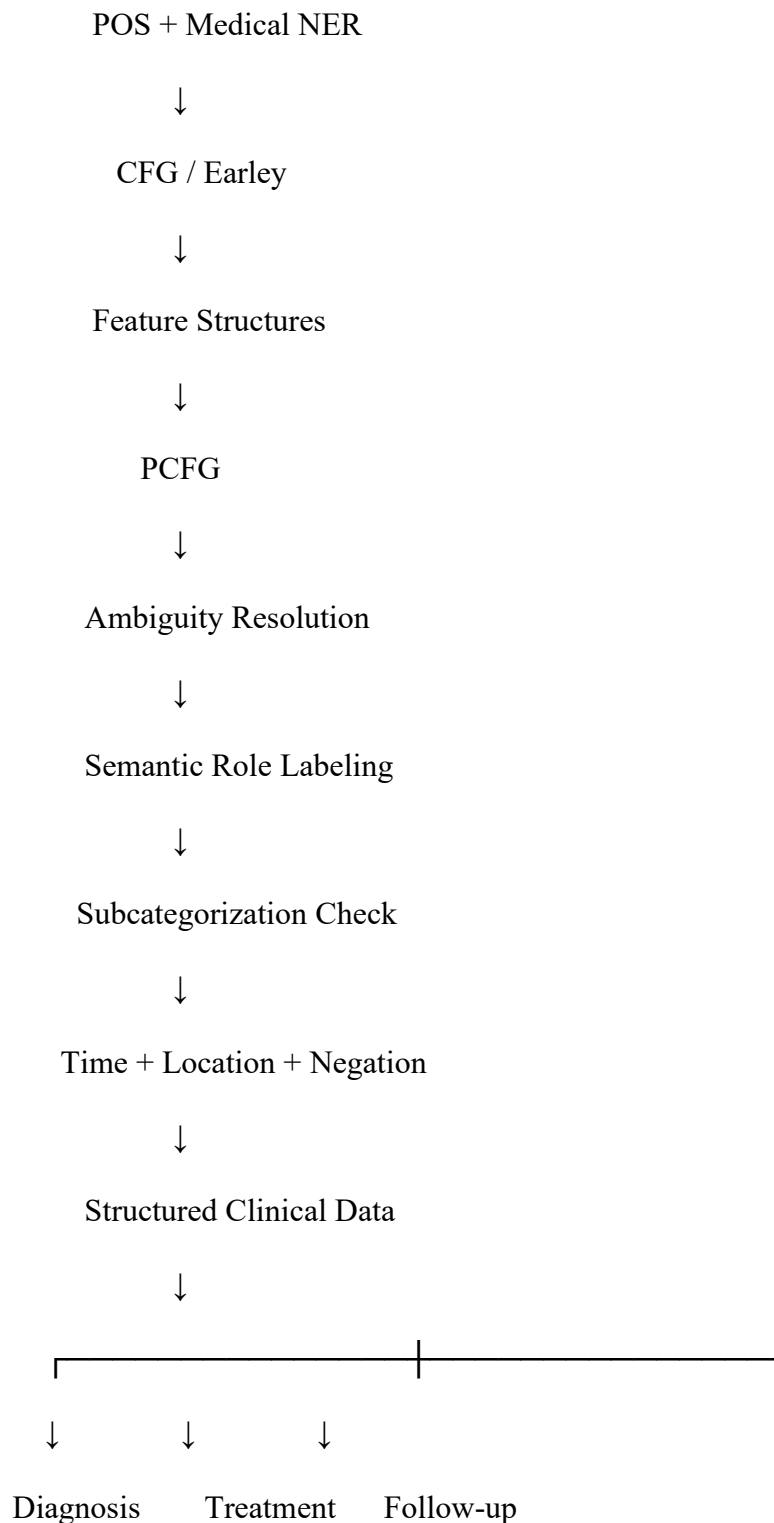
For a hospital environment processing thousands of reports, the system should use:

- modular NLP components;
- efficient chart parsing;
- reusable grammar rules;
- medical terminology databases;
- parallel processing;
- confidence-based validation;
- scalable cloud/server infrastructure.

Final proposed system

Medical Reports





Conclusion

A basic CFG alone is insufficient for complex healthcare reports because it cannot effectively resolve ambiguity, enforce detailed agreement constraints, or represent medical relationships. A **hybrid NLP architecture combining CFG, PCFG, feature structures, efficient parsing, semantic-role labeling, medical NER, and subcategorization frames** provides more accurate, scalable, and reliable clinical information extraction.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____